

integral-1

$$(3\sec^2 2x)' = \frac{3 \tan 2x}{2}$$

$\sec^2 2x$

① $\int (\cos 4x + \sin 2x - 2x^3 + \frac{3}{\cos^2 2x} + 2) dx = ?$

$$\frac{\sin 4x}{4} - \frac{\cos 2x}{2} - \frac{2x^4}{4} + \frac{3 \tan 2x}{2} + 2x + c$$

② $\int (\frac{2}{1+x^2} - \frac{3}{\sqrt{1-x^2}} + \frac{1}{x} - \frac{2}{x^2} + e^{4x} + x + x^2) dx = ?$

$$2 \arctan x - 3 \arcsin x + \ln|x| + \frac{2}{x} + \frac{e^{4x}}{4} + \frac{x^2}{2} + \frac{x^3}{3} + c$$

③ $\int x^2 \cdot \cos(x^3+2) dx = ?$

$$x^3+2=u \quad 3x^2 dx = du$$

$$\int \frac{\cos u}{3} du = \frac{\sin u}{3} + c$$

$$\frac{\sin(x^3+2)}{3} + c$$

④ $\int_0^2 \frac{dx}{\sqrt{x+2} \cdot \cos^2 \sqrt{x+2}} = ?$

$$\sqrt{x+2} = u$$

$$\frac{1}{2\sqrt{x+2}} dx = du$$

$$\int_0^2 \frac{2}{\cos^2 u} du = \int_0^2 2 \sec^2 u du = 2 \tan u \Big|_0^2 = 2 \tan \sqrt{x+2} \Big|_0^2 = 2 \tan 2 - 2 \tan \sqrt{2}$$

⑤ $\int (\tan 4x + \cot 3x) dx = ?$

$$-\frac{\ln|\cos 4x|}{4} + \frac{\ln|\sin 3x|}{3} + c$$

$$\int \tan x dx = -\ln|\cos x| + c$$

$$\int \cot x dx = \ln|\sin x| + c$$

⑥ $\int_0^{\pi/2} \cos x \cdot e^{2\sin x+3} dx = ?$

$$2\sin x+3=u$$

$$2\cos x dx = du$$

$$\frac{1}{2} \int_0^{\pi/2} e^u du = \frac{1}{2} (e^{2\sin x+3} \Big|_0^{\pi/2})$$

$$\frac{1}{2} (e^5 - e^3)$$

⑦ $\int_0^1 \frac{x+1}{\cos(x^2+2x+1)} dx = ?$

$$x^2+2x+1=u$$

$$2x+2 dx = du$$

$$\frac{1}{2} \int_0^1 \frac{1}{\cos u} du = \frac{1}{2} \int_0^1 \sec u du = \frac{1}{2} (\ln|\sec u + \tan u| \Big|_0^1)$$

$$\frac{1}{2} (\ln|\sec 4 + \tan 4| - \ln|\sec 1 + \tan 1|)$$

⑧ $\int \frac{dx}{x \cdot \tan(\ln x)} = ?$

$$\ln x = u$$

$$\frac{1}{x} dx = du$$

$$\int \cot u du = \ln|\sin u| + c = \ln|\sin(\ln x)| + c$$

⑨ $\int_0^{\pi/4} \frac{e^{\tan x}}{\cos^2 x} dx = ?$

$$\tan x - 2 \ln(\cos x)$$

$$e^{\tan x} = u$$

$$e^{\tan x} \cdot \sec^2 x dx = du$$

$\frac{e-1}{2}$
soruldugu sene gok az agrencinin dogru cozebildigi basit bir soru eslinde

Cevaplar

$$\textcircled{1} \int \left(\cos 4x + \sin 2x - 2x^3 + \frac{3 \sec^2 2x}{\cos^2 2x} + 2 \right) dx = ?$$

$$= \frac{\sin 4x}{4} - \frac{\cos 2x}{2} - 2 \cdot \frac{x^4}{4} + 3 \cdot \frac{\tan 2x}{2} + 2x + C$$

$$\textcircled{2} \int \left(\frac{2}{1+x^2} - \frac{3}{\sqrt{1-x^2}} + \frac{1}{x} - \frac{2}{x^2} + e^{4x} + x + x^2 \right) dx = ?$$

$$= \underbrace{2 \operatorname{Arctan} x}_{\text{Bunun yerine}} - \underbrace{3 \operatorname{ArcSin} x}_{\text{Bunun yerine}} + \ln|x| + \frac{2}{x} + \frac{e^{4x}}{4} + \frac{x^2}{2} + \frac{x^3}{3} + C$$

Bunun yerine $-2 \operatorname{Arccot} x$ de yazabilirsiniz
Bunun yerine $+3 \operatorname{Arccos} x$ yazabilirsiniz

Türevi burda 😊

$$\textcircled{3} \int x^2 \cdot \cos(x^3+2) dx = ?$$

$$u = x^3 + 2 \\ du = 3x^2 dx \rightarrow x^2 dx = \frac{du}{3}$$

$$= \int \frac{1}{3} \cdot \cos u \, du = \frac{1}{3} \sin u + C = \frac{1}{3} \sin(x^3+2) + C$$

$$\textcircled{4} \int_0^2 \frac{dx}{\sqrt{x+2} \cdot \cos^2 \sqrt{x+2}} = ?$$

$$u = \sqrt{x+2} \quad du = \frac{1}{2\sqrt{x+2}} dx$$

$$u = \sqrt{x+2}$$

$$\begin{array}{lcl} \text{E.S} & \rightarrow & \text{Y.S} \\ x=0 & \rightarrow & u=\sqrt{2} \\ x=2 & \rightarrow & u=2 \end{array} \left| \frac{dx}{\sqrt{x+2}} = 2 du \right.$$

$$= \int_{\sqrt{2}}^2 \frac{2 du}{\cos^2 u}$$

$$= 2 \tan u \Big|_{\sqrt{2}}^2 = 2 \tan 2 - 2 \tan \sqrt{2}$$

$$\textcircled{5} \int (\tan 4x + \cot 3x) dx = ?$$

$$\int \tan x dx = -\ln |\cos x| + c$$

$$\int \cot x dx = \ln |\sin x| + c$$

$$= -\frac{1}{4} \ln |\cos 4x| + \frac{1}{3} \ln |\sin x| + c$$

$$\textcircled{6} \int_0^{\pi/2} \cos x \cdot e^{2\sin x + 3} dx \Rightarrow$$

Törvi burda

$$u = 2\sin x + 3 \quad du = 2\cos x dx$$

$$\downarrow$$

$$\cos x dx = \frac{du}{2}$$

E.S. 4.S

$$u=0 \rightarrow u=3$$

$$u=\frac{\pi}{2} \rightarrow u=5$$

$$= \int_3^5 \frac{1}{2} \cdot e^u du = \frac{1}{2} e^u \Big|_3^5 = \frac{1}{2} (e^5 - e^3)$$

$$\textcircled{7} \int_0^1 \frac{x+1}{\cos(x^2+2x+1)} dx$$

$$x^2 + 2x + 1 = u \rightarrow du = (2x+2) dx$$

$$\downarrow$$

E.S. 4.S

$$x=0 \rightarrow u=1$$

$$x=1 \rightarrow u=4$$

$$\frac{du}{2} = (x+1) dx$$

$$= \int_1^4 \frac{1}{2} \cdot \frac{du}{\cos u} = \frac{1}{2} \int_1^4 \sec u du = \frac{1}{2} \ln |\sec u + \tan u| \Big|_1^4$$

$$= \frac{1}{2} \ln |\sec 4 + \tan 4| - \frac{1}{2} \ln |\sec 1 + \tan 1|$$

$$\textcircled{8} \int \frac{dx}{x \cdot \tan(\ln x)} = ? \quad u = \ln x \quad du = \frac{dx}{x}$$

$$= \int \frac{du}{\tan u} = \int \cot u \, du = \ln |\sin u| + C = \ln |\sin(\ln x)| + C$$

$$\textcircled{9} I = \int_0^{\pi/4} e^{\tan x - 2 \ln(\cos x)} dx = ?$$

$$* (e^{\ln f(x)} = f(x)) *$$

$$\begin{aligned} e^{\tan x - 2 \ln(\cos x)} &= e^{\tan x} \cdot e^{-2 \ln(\cos x)} = e^{\tan x} \cdot e^{\ln(\cos x)^{-2}} \\ &= e^{\tan x} \cdot (\cos x)^{-2} \\ &= e^{\tan x} \cdot \sec^2 x \end{aligned}$$

$$I = \int_0^{\pi/4} \underbrace{e^{\tan x}}_{e^u} \cdot \underbrace{\sec^2 x dx}_{du}$$

$$\begin{aligned} &= \int_0^1 e^u du = e^u \Big|_0^1 \\ &= e - 1 \end{aligned}$$

$$u = \tan x \rightarrow du = \sec^2 x dx$$

<u>E.S</u>		<u>Y.S</u>
$x = 0$	$\longrightarrow$	$u = 0$
$x = \frac{\pi}{4}$	$\longrightarrow$	$u = 1$